

Overall System Architecture

Five-layer greenhouse harvesting pipeline

Layer 1 - Simulation and Robot Platform

Role: Gazebo greenhouse world, robot body, crops, baskets, and sensor models
Main elements: Panther base, Franka FR3 arm, RGB-D camera, LiDAR, IMU, B/C tomato rows

simulation data

Layer 2 - ROS2 Interface and Robot State

Role: Bridge simulation data into ROS2 and expose robot commands/state
Main modules: ros_gz_bridge, ros2_control, robot_state_publisher, TF2 tree

scan, odom, TF

RGB-D + TF

Layer 3 - Navigation and Localization

Role: Localize the base on the map and follow plant-row goals
Main modules: SLAM Toolbox, AMCL, robot_localization EKF, Nav2

pose + waypoints

Layer 4 - Perception and Tomato Mapping

Role: Detect tomatoes, estimate 3D positions, and publish a live tomato map
Main modules: YOLO detector, TomatoDepthMapper, /tomato_map/list, bbox viewer, map panel

tomato map

motion commands

Layer 5 - Mission, Planning, and Harvesting

Role: Select targets, avoid obstacles, and execute pick-place motions
Main modules: MissionManager, collision scene, target selector, MoveIt/OMPL, Pilz LIN

Closed-loop execution: commands return to controllers; Gazebo attach/detach updates tomato state.